

Coverage audit: how the four implementations cover the fragment's constructions

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1 WHAT THIS AUDIT IS

The article's §5 compares four implementations of the same facts: the D-noun categorization with ordinary Head (I₁), a separate determinative category with ordinary Head (I₂), a separate category with *CGEL*'s fusion of functions (I₃), and the D-noun categorization with fusion (I₄). It states which conditions the accounts share and which are specific to one of them. This audit records, for each construction family the fragment covers, and for each lexeme group the article distinguishes, it records how each implementation covers the facts: by a general rule of that grammar (D), by a general rule that applies only because the grammar admits a determinative as the lexical head of Nom and NP (D_n), by a rule the grammar has to state once for nouns and again for determinatives, or once for NP and again for DP (D₂), by a fact stored on the lexeme (L), by a condition stated for that construction alone (S), or not at all (U).

The classifications are the author's, made by hand from the fragment's stated conditions and the sections cited beside each cell; they are the analysis, not data. The precedence rule is this: a cell records the mechanism by which the implementation covers the licensed use. Lexical permissions, target selection, forms, and inflection are held fixed across the four implementations and are coded L only where the fact is nothing but such a permission or exclusion; where a licensed use is covered by a rule, the rule's class is recorded even though the permission is presupposed. S and D₂ codes count cells; the distinct conditions and duplicated rules behind them are listed separately (Table 5), since a condition stated once may apply in several cells. The accounts are compared as complete statements of the same fragment, over the rules and facts each has to state, and not by the number of categories each posits; that is the standard the minimum description length principle makes explicit (Rissanen, 1978). Beside each cell the audit prints the number of participation claims the claim register records for that family and group, as licensed/conditional/excluded, so that a cell's classification can be checked against what the article asserts there. Cells with no claim and no analysis in the article are left out; the article's analysis, not the register, decides which cells exist.

2 THE FOUR IMPLEMENTATIONS AND THEIR PRIMITIVES

The classes count rules, not primitives. The primitives differ too, and Table 1 lists them, since a rule that is general in one grammar may be general only because that grammar has a primitive the other lacks.

Table 1: Primitives of the four implementations.

	I ₁	I ₂	I ₃	I ₄
Lexical categories	Noun, four subcategories	Noun (three) and D	Noun (three) and D	Noun, four subcategories
Category condition on lexical Heads in Nom	N	N or D	N or D	N
Phrase types in Det	NP, PP	NP, PP	DP, NP, PP	NP, PP
Head configurations for independent determinatives	ordinary Head	ordinary Head	Det–Head and Mod–Head fusion	Det–Head and Mod–Head fusion
Adjectival Mod–Head fusion (<i>the rich</i>)	yes	yes	yes	yes
Restrictions stated on the fused inner phrase	none	none	yes	yes
Complex determinatives listed	none	none	<i>a few, a little, many a, a great many</i>	none
Conversion rule for the set-denoting use	none	yes	yes	none
Domain of the head-genitive definition	Noun heads	Noun and D heads	Noun and D heads	Noun heads

3 RESULTS

Table 2: Cells by class, over the 103 classified cells.

Implementation	D	Dn	D ₂	L	S	U
I ₁ : D-noun, ordinary Head	61	0	0	36	6	0
I ₂ : Separate D, ordinary Head	28	32	1	36	6	0
I ₃ : Separate D, fused Head	40	0	12	35	16	0
I ₄ : D-noun, fused Head	53	0	0	36	14	0

Table 3: Cells on which the implementations differ.

Family	Lexeme group	I ₁	I ₂	I ₃	I ₄
independent argument use, including the existential	no/none	D	Dn	D	D
independent argument use, including the existential	demonstratives, such, and the wh-forms	D	Dn	D	D
independent argument use, including the existential	the general quantifiers	D	Dn	D	D

Family	Lexeme group	I ₁	I ₂	I ₃	I ₄
independent argument use, including the existential	the gradable four	D	Dn	D	D
independent argument use, including the existential	cardinals	D	Dn	D	D
independent argument use, including the existential	compound determinatives	D	Dn	D	D
independent use with a partitive of-PP	no/none	D	Dn	S	S
independent use with a partitive of-PP	the general quantifiers	D	Dn	S	S
independent use with a partitive of-PP	the gradable four	D	Dn	S	S
independent use with a partitive of-PP	cardinals	D	Dn	S	S
external determination of an independent use	the gradable four	D	Dn	D	D
external determination of an independent use	cardinals	D	Dn	D	D
internal modifiers admitted	the gradable four	D	Dn	D	D
peripheral modifiers admitted	every	D	Dn	D ₂	D
peripheral modifiers admitted	no/none	D	Dn	D ₂	D
peripheral modifiers admitted	the general quantifiers	D	Dn	D ₂	D
peripheral modifiers admitted	cardinals	D	Dn	D ₂	D
peripheral modifiers admitted	compound determinatives	D	Dn	D ₂	D
relative postmodification	demonstratives, such, and the wh-forms	D	Dn	S	S
relative postmodification	the general quantifiers	D	Dn	S	S
relative postmodification	the gradable four	D	Dn	S	S
relative postmodification	cardinals	D	Dn	S	S
relative postmodification	compound determinatives	D	Dn	S	S
number agreement and transparency	my/mine	S	S	D	D
number agreement and transparency	the general quantifiers	D	Dn	D ₂	D
number agreement and transparency	the gradable four	D	Dn	D ₂	D
genitive marking	compound determinatives	D	D ₂	D ₂	D
degree modification outside the NP; comparative complements; predicative use	the articles	D	Dn	D ₂	D
degree modification outside the NP; comparative complements; predicative use	no/none	D	Dn	D ₂	D

Family	Lexeme group	I ₁	I ₂	I ₃	I ₄
degree modification outside the NP; comparative complements; predicative use	the general quantifiers	D	D _n	D ₂	D
degree modification outside the NP; comparative complements; predicative use	the gradable four	D	D _n	D ₂	D
predeterminer modification and the former complex determinatives	the articles	L	L	S	L
predeterminer modification and the former complex determinatives	demonstratives, such, and the wh-forms	D	D _n	D	D
predeterminer modification and the former complex determinatives	the general quantifiers	D	D _n	D	D
predeterminer modification and the former complex determinatives	the gradable four	D	D _n	S	D

Table 4: What each implementation states for a construction alone (S) or states twice (D₂).

Account	Stated for the construction alone	Stated twice
I ₁	internal modifiers (cardinals); the compound construction (compound determinatives); number agreement (my/mine); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	–
I ₂	internal modifiers (cardinals); the compound construction (compound determinatives); number agreement (my/mine); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	genitive marking (compound determinatives)
I ₃	partitive of-PP (no/none, the general quantifiers, the gradable four, cardinals); internal modifiers (cardinals); relative postmodification (demonstratives, such, and the wh-forms, the general quantifiers, the gradable four, cardinals, compound determinatives); the compound construction (compound determinatives); predeterminer modification and the former complex determinatives (the articles, the gradable four); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	peripheral modifiers (every, no/none, the general quantifiers, cardinals, compound determinatives); number agreement (the general quantifiers, the gradable four); genitive marking (compound determinatives); degree use outside the NP (the articles, no/none, the general quantifiers, the gradable four)
I ₄	partitive of-PP (no/none, the general quantifiers, the gradable four, cardinals); internal modifiers (cardinals); relative postmodification (demonstratives, such, and the wh-forms, the general quantifiers, the gradable four, cardinals, compound determinatives); the compound construction (compound determinatives); the secondary set-denoting use (cardinals, compound determinatives, proper nouns)	–

Table 5: Distinct conditions stated for a construction alone, and distinct rules stated twice, per implementation. One condition may cover several cells.

Implementation	Conditions	Rules stated twice	The conditions
I1	4	0	cardinal two sites, compound restrictor, mine agreement, secondary use rule
I2	5	1	cardinal two sites, compound restrictor, conversion rule, mine agreement, secondary use rule
I3	6	4	cardinal two sites, complex listing, compound restrictor, conversion rule, inner phrase restriction, secondary use rule
I4	4	0	cardinal two sites, compound restrictor, inner phrase restriction, secondary use rule

Table 6: Sensitivity: cells by class under the report’s principles (1)–(4).

Implementation	D	Dn	D2	L	S	U
I1	54	0	0	43	6	0
I2	29	25	0	43	6	0
I3	54	0	0	42	7	0
I4	55	0	0	43	5	0

Table 7: Sensitivity: cells by class under the same with the disjunct priced at zero (5).

Implementation	D	Dn	D2	L	S	U
I1	54	0	0	43	6	0
I2	54	0	0	43	6	0
I3	54	0	0	42	7	0
I4	55	0	0	43	5	0

Fusion is held fixed for adjectives (*the rich*) in all four implementations. *CGEL* (I3) uses it a second time, for independent determinatives, and pays for that use with one condition on the fused inner phrase (nine cells) and with rules stated once for NPs and again for DPs (twelve cells, four rules). The two ordinary-Head implementations remove the second use and share the saving; they buy it differently. D-noun (I1) moves the determinatives one node down the taxonomy and keeps the head condition as it is. Separate D with ordinary Head (I2) keeps D a primary category and adds a disjunct to the head condition, and the Dn column records the 32 derivations that go through that disjunct. For the same coverage, I2 therefore carries what I1 carries plus the disjunct, plus a primary category whose defining profile duplicates Noun’s (the nominal connections of the article’s §2), plus the conversion rule and the extended genitive definition (Table 5). A cross-cutting nominal feature wouldn’t change the count. A feature carried by exactly the nouns and the determinatives has the extension of the superordinate node, and it costs its definition and one assignment per lexeme where the node

costs itself and its inheritance links; a feature would earn its place by cross-cutting the categories, and in this fragment none does. The lexical facts (use permissions, target selection, forms, inflection) are the same in all four.

The sensitivity tables apply the second-round report's alternative principles: lexical licensing first, the inner-phrase restriction counted once, typed schemas in place of duplicated rules, the secondary use as a rule in every account. Under those principles I₁, I₃, and I₄ come out within one cell of each other and I₂ trails only by its D_n cells; pricing the disjunct at zero brings I₂ level too. That is where the disagreement lies: the recoding prices the disjunct at zero and doesn't count the category it keeps. The distinct-conditions table counts both, and there I₂ is I₁ plus one condition and one category. The article's §5.5 states the comparison in these terms.

4 THE GRID

Table 8: The grid. Codes: D derived by a general rule; D_n derived only by admitting D as the head of Nom and NP; D₂ derived by a rule stated once for nouns and again for determinatives (or for NP and DP); L stored on the lexeme; S stated for this construction alone; U not covered. Claims: register claims for the cell, as licensed/conditional/excluded.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
<i>Det before a nominal (§2.1)</i>						
the articles	L	L	L	L	8/2/1	Det use permission and target selection (a: singular count) are lexical in every account; the phrase in Det is an NP under I ₁ , I ₂ , and I ₄ and a DP under I ₃ (primitives table).
every	L	L	L	L	4/1/1	as the articles; the Mod use after a genitive Det is a separate lexical permission (internal Mod family).
no/none	L	L	L	L	1/1/1	the dependent form no is selected by the Det use (forms family).
my/mine	L	L	L	L	1/2/1	a genitive NP in Det, the general realization; the my/mine form selection is lexical (forms family).
demonstratives, such, and the wh-forms	L	L	L	L	4/1/0	Det use permission; agreement with the target is the demonstratives' inflection (inflection family).
the general quantifiers	L	L	L	L	8/1/1	Det use permission with target restrictions (some: plural or non-count; each: singular count).
the gradable four	L	L	L	L	4/1/1	Det use permission with target restrictions (few: plural count; much: non-count).
cardinals	L	L	L	L	3/1/0	Det use with plural targets above one; complex cardinals fill Det as phrases (§5.4).
compound determinatives	D	D	D	D	1/0/0	only as genitive NPs (someone's preferences): the general genitive-Det realization.
pronouns	D	D	D	D	0/0/2	genitive NPs (my preferences, her preferences) fill Det by the general genitive-Det realization; plain-case pronouns don't.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	2/o/o	genitive NPs (people's preferences) by the general realization; plain-case NPs (what size shoes, Sunday morning) by a restricted construction that every account states separately.
proper nouns	D	D	D	D	5/o/o	genitive NPs (Kim's preferences); Sunday morning is the restricted plain-case construction.
<i>internal Mod before a nominal (§4.3)</i>						
every	L	L	L	L	1/1/o	Mod after a genitive Det (her every move): a lexical permission.
the general quantifiers	L	L	L	L	1/o/1	several, certain, various as modifiers (its several advantages): lexical.
the gradable four	L	L	L	L	4/1/o	the few people, her many virtues: a lexical permission for Mod use; CGEL gives few Mod after a definite determiner (p. 392).
cardinals	L	L	L	L	2/o/o	these two hundred books, the remaining three: Mod use of cardinals.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	2/o/o	nominal premodifiers (dog houses): general.
proper nouns	D	D	D	D	1/o/o	Canada Day: general nominal premodification, restricted for names.
adjective controls	D	D	D	D	4/o/o	attributive AdjPs: general.
<i>independent argument use, including the existential (§4.2)</i>						
the articles	L	L	L	L	1/1/6	no independent argument use: the lexeme lacks the use permission in every account.
every	L	L	L	L	1/o/5	excluded by the lexeme's permissions in every account; the corpus check is tabled in the quantifier supplement.
no/none	D	Dn	D	D	4/1/2	none as the independent form (forms family); the use is licensed by the ordinary Head chain (I ₁ , I ₂) or by Det–Head fusion (I ₃ , I ₄). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
my/mine	D	D	D	D	2/o/2	mine: ordinary Head (I ₁ , I ₂) or fusion with an understood nominal (I ₃ , I ₄); the agreement condition is in the agreement family.
demonstratives, such, and the wh-forms	D	Dn	D	D	7/o/1	independent this/that; which and what as interrogative heads. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	D	D	19/8/o	some, all, both, each, several, enough independently, and the existential There are several: ordinary Head (I ₁ , I ₂) or Det–Head fusion (I ₃ , I ₄). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
the gradable four	D	Dn	D	D	11/3/0	few and many independently, including bare few with no set under discussion (§4.2). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	D	D	1/0/0	independent two, ten; the almost thirty who came is in the external-determination family. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	D	Dn	D	D	4/0/0	someone, anyone as NPs; under I ₃ the DP–Nom structure of Payne 2007. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
pronouns	D	D	D	D	11/0/2	the noun-headed NP without a determiner (She left).
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	6/1/1	bare plurals and non-counts; singular count arguments require determination, a condition every account states.
proper nouns	D	D	D	D	5/0/0	primary naming uses (Kim left).
adjective controls	D	D	D	D	8/4/0	the rich by adjectival Mod–Head fusion, retained in all four; bare numerous, multiple, countless are attested overlaps the article leaves unanalysed (§2.3), so a U would be right for them alone.
<i>independent use with a partitive of-PP (§4.3)</i>						
every	L	L	L	L	0/0/1	no partitive use (each of, not every of): lexical.
no/none	D	Dn	S	S	0/2/1	none of the students: the of-PP is a complement in the Nom that none heads (I ₁ , I ₂); under fusion it attaches outside the DP, which the inner-phrase restriction has to state (§5.2). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	S	S	3/3/0	some of the wine, many of them: complement in the Nom the determinative heads (I ₁ , I ₂); outside the fused DP by a stated restriction (I ₃ , I ₄), §5.2. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	Dn	S	S	2/1/1	few of them, many of the delegates: as for the general quantifiers. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	S	S	–	two of them: as above. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	4/0/0	a lot of the wine, a number of protesters: complement of the head noun in every account (CGEL's own analysis of a number of, pp. 351–352).
<i>external determination of an independent use (§4.3)</i>						
the articles	L	L	L	L	1/0/0	excluded: the permissions of independent few don't transfer to the articles (§5.2).
every	L	L	L	L	–	excluded, as the articles.
the general quantifiers	L	L	L	L	–	excluded (the some, the all): lexical.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
every	D	Dn	D ₂	D	4/o/o	almost every: NP-peripheral attachment by the general rule for peripheral modifiers, with lexical selection (I ₁ , I ₂ , I ₄); a modifier inside the DP under CGEL, so the same fact needs the DP-modifier rule beside the NP-peripheral rule that covers only you (I ₃), §4.4. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
no/none	D	Dn	D ₂	D	3/o/o	practically no: as for every. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	D ₂	D	5/o/o	nearly all, hardly any: as for every. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	D ₂	D	1/o/o	almost ten: as for every. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	D	Dn	D ₂	D	–	hardly anyone: peripheral to the NP that anyone heads (I ₁ , I ₂ , I ₄); in DP structure under Payne 2007 (I ₃). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
pronouns	D	D	D	D	2/o/o	only you, all we who signed up: NP-peripheral in every account.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	–	only the book, almost the whole class.
proper nouns	D	D	D	D	–	even Kim.
<i>relative postmodification</i> (§2.1)						
demonstratives, such, and the wh-forms	D	Dn	S	S	–	those who came, that which remains: as for few. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	Dn	S	S	–	some that I saw: as above. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	Dn	S	S	–	few who come ever leave: an ordinary postmodifier of the Nom that few heads (I ₁ , I ₂); attaches outside the fused DP, which the inner-phrase restriction states (I ₃ , I ₄). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
cardinals	D	Dn	S	S	–	two that I have seen: as above. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
compound determinatives	D	Dn	S	S	–	anyone who asks, something that you need to know: as above, and ordered after the restrictor (compound family). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
pronouns	L	L	L	L	–	we who have read the report: a restricted range, lexical.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	–	people who came: general.
<i>the compound construction: base permissions and the post-head restrictor (§4.5)</i>						
the general quantifiers	D	D	D	D	1/o/o	the base supplies the compound's premodifier permissions (almost anybody from almost any): inheritance from the base in every account.
compound determinatives	S	S	S	S	4/o/o	the post-head restrictor's position and non-recursion, the exclusion of external determination, and the exclusion of a pre-head adjective are conditions of the compound construction in every account (§4.5); the premodifier permissions inherited from the base (hardly any, hardly anyone) are general in all four.
<i>dependent and independent forms (no/none, my/mine) (§3.2)</i>						
no/none	L	L	L	L	–	inflectional forms of one lexeme selected by the use; the partitive requires none: lexical.
my/mine	L	L	L	L	–	my/mine: as no/none.
pronouns	L	L	L	L	–	she/her: case selected by function.
<i>number agreement and transparency (§2.2)</i>						
my/mine	S	S	D	D	2/o/o	Mine is ready / Mine are ready: ordinary Head needs a constructional agreement condition (§4.6); fusion takes the number from the understood nominal (I ₃ , I ₄).
demonstratives, such, and the wh-forms	L	L	L	L	–	this/these agree with the target: the lexeme's inflection.
the general quantifiers	D	D _n	D ₂	D	1/o/o	some of the people were: the transparency rule for quantificational heads, stated once over Nom heads (I ₁) or once over {N, D} heads (I ₂); CGEL states it for the number-transparent nouns and again for the fused-head partitive (I ₃); under I ₄ the rule for nouns covers some. each of the people takes singular agreement by default, a lexical fact in every account. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	D _n	D ₂	D	–	many of them were; a few takes plural agreement in fused-head use: as for the general quantifiers. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	3/o/2	a lot of the work was, a lot of the people were: the number-transparent nouns (CGEL §5.3.3), in every account.
<i>number and grade inflection (§2.5)</i>						
demonstratives, such, and the wh-forms	L	L	L	L	2/o/o	number inflection of two lexemes: lexical in every account; the paradigm parallels nominal number.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
the gradable four	L	L	L	L	8/o/o	grade inflection (few/fewer/fewest): lexical; shared with adjectives.
cardinals	L	L	L	L	–	plural cardinals (in threes) are the secondary use (below).
pronouns	L	L	L	L	o/1/o	case and reflexive forms.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	–	count nouns inflect for number: a general rule with lexical exceptions.
proper nouns	L	L	L	L	–	restricted number (the Smiths) is the secondary use (below).
<i>genitive marking (§2.5)</i>						
compound determinatives	D	D ₂	D ₂	D	1/2/o	everyone's, everyone else's: the head-genitive definition, 'the inflection of the head noun' (CGEL p. 479), covers a compound within Noun (I ₁ , I ₄); with a separate D the definition has to be extended to D heads (I ₂ , I ₃), §2.5.
pronouns	L	L	L	L	–	my/mine, her: suppletive genitive forms.
common nouns, including quantificational lot, plenty, heaps	D	D	D	D	1/o/o	dog's, the King of England's: head and phrasal genitives.
proper nouns	D	D	D	D	1/o/o	Kim's, Edward's.
<i>degree modification outside the NP; comparative complements; predicative use (§5.3)</i>						
the articles	D	D _n	D ₂	D	2/o/1	the bigger the better: the as a degree modifier of comparative AdjPs, as much bigger and no better; an NP modifier (I ₁ , I ₂ , I ₄) or a DP modifier (I ₃). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
no/none	D	D _n	D ₂	D	1/o/o	no better: as the bigger. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	D _n	D ₂	D	6/1/o	enough modifying adjectives, adverbs, verbs, and PPs (good enough, quickly enough): an NP modifier under the three NP-projecting implementations, under the permission that covers a great deal smaller and lots better (I ₁ , I ₂ , I ₄); a DP modifier that needs its own permission beside the NP permission (I ₃), §5.3. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	D _n	D ₂	D	4/1/o	much bigger, a lot fewer: as for enough; the comparative complement in more than ten is licensed separately in every account (lexical), and predicative Their enemies were many is a restricted lexical fact. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.

Lexeme group	I ₁	I ₂	I ₃	I ₄	Claims	Basis
common nouns, including quantificational lot, plenty, heaps <i>predeterminer modification and the former complex determinatives (§3.1)</i>	D	D	D	D	13/o/1	a great deal smaller, lots better, heaps worse, three years old: NP modifiers of adjectives in every account (CGEL pp. 549–550).
the articles	L	L	S	L	–	a in a few, many a, such a: the article's ordinary Det use with count-singular selection (I ₁ , I ₂ , I ₄); part of a listed unit under I ₃ .
demonstratives, such, and the wh-forms	D	D _n	D	D	–	such a mess, what a mess: degree such and exclamative what in predeterminer function (I ₁ , I ₂ , I ₄); adjectives in the same function under CGEL (I ₃). The difference is the category, not the machinery. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the general quantifiers	D	D _n	D	D	2/o/o	all the books, both the copies: a predeterminer modifier realized by the determinative's phrase in every account (a DP in CGEL, an NP here). Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
the gradable four	D	D _n	S	D	o/3/o	a few, a little, a great many: an NP in Det with its own Det, as genitive Dets already are (I ₁ , I ₂ , I ₄); many a: many in predeterminer function on an a-NP. CGEL lists a few, a little, many a, and a great many as complex determinatives with a sui generis internal structure (I ₃), §3.1. Under I ₂ the derivation goes through the admission of D as a head of Nom and NP.
common nouns, including quantificational lot, plenty, heaps <i>the secondary set-denoting use (§4.5)</i>	D	D	D	D	1/o/o	half a cake, and the fractions: nouns and NPs in predeterminer function in every account (CGEL p. 385 n. 25).
cardinals	S	S	S	S	–	in threes, in their hundreds: the same rule (I ₁ , I ₄); CGEL's 'secondary use in which they inflect for number and hence belong in the noun category' is a change of category, a second rule (I ₂ , I ₃), §5.5.
compound determinatives	S	S	S	S	–	a special someone, such an anyone, the somebodies: the secondary-use rule of §4.5, one rule over Noun (I ₁ , I ₄); the same rule for nouns plus a conversion rule from D (I ₂ , I ₃ ; CGEL p. 423 n. 43), §4.5 and §5.5.
proper nouns	S	S	S	S	–	an Ophelia, the Smiths: CGEL's own secondary use within Noun (p. 521), the same rule in every account.

5 LIMITS

The grid covers the fragment's constructions and nothing outside them: predication, interrogative clause structure, coordination, and modification outside NP structure are not classified, as the article's §5.1 says of the fragment itself. The classes are coarse by design; a D under one implementation and a D under another can rest on different primitives (Table 1), and the summary counts

should be read with that table. The classifications are contestable at the cell, and the JSON file (analysis/coverage-audit.json) is written so that a reader can change one cell and regenerate the tables (analysis/tools/coverage_audit.py).

REFERENCES

Rissanen, J. (1978). Modeling by shortest data description. *Automatica*, 14(5), 465–471. [https://doi.org/10.1016/0005-1098\(78\)90005-5](https://doi.org/10.1016/0005-1098(78)90005-5)